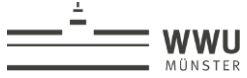


# Business Process Management

## Introduction





## Christian's CV





# Your Expectations?



<http://www.nichtlustig.de/>

# Code of Conduct



- Please ask **questions** if something is unclear or if there is a discussion



- Please use the platform **moodle**, to ask questions, download material and hand in assignments



- Please prepare the lectures and assignments, by looking at and reading the material **prior to coming to the lecture**



- Please **do not distribute** the material provided to you in this course

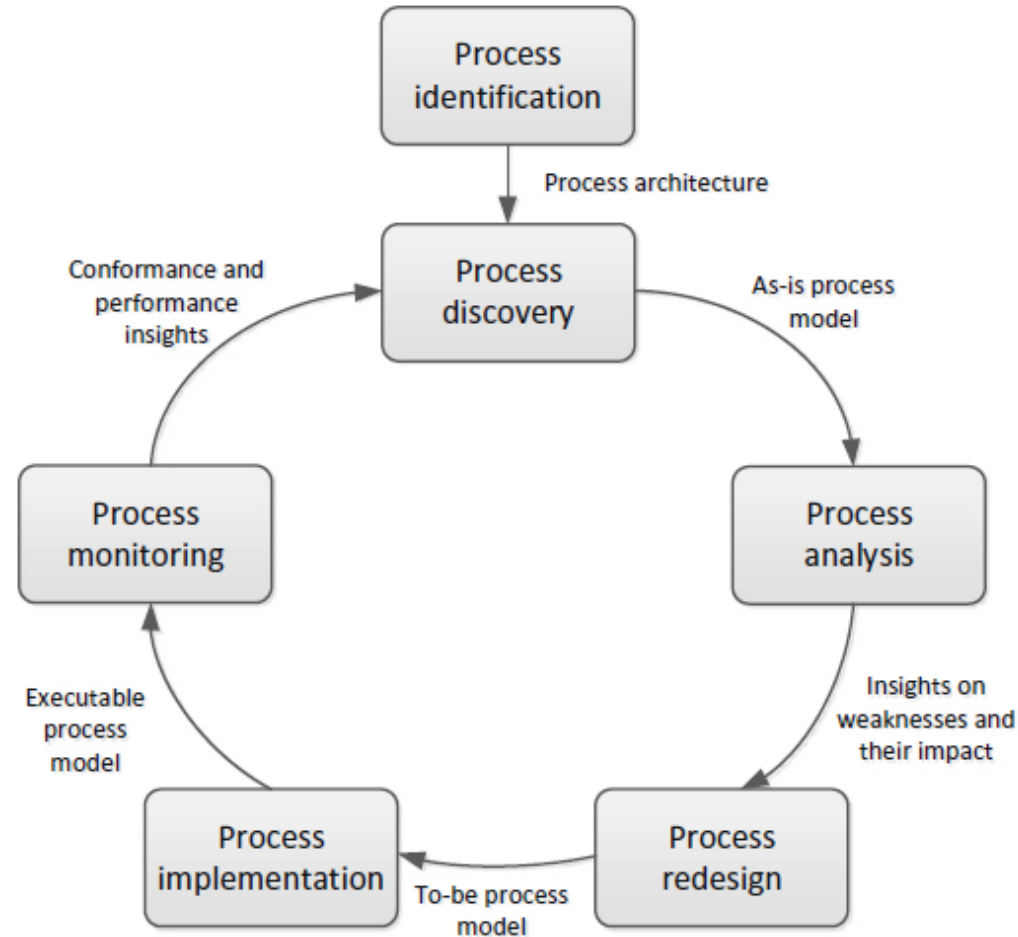
# Let's Get Organized!



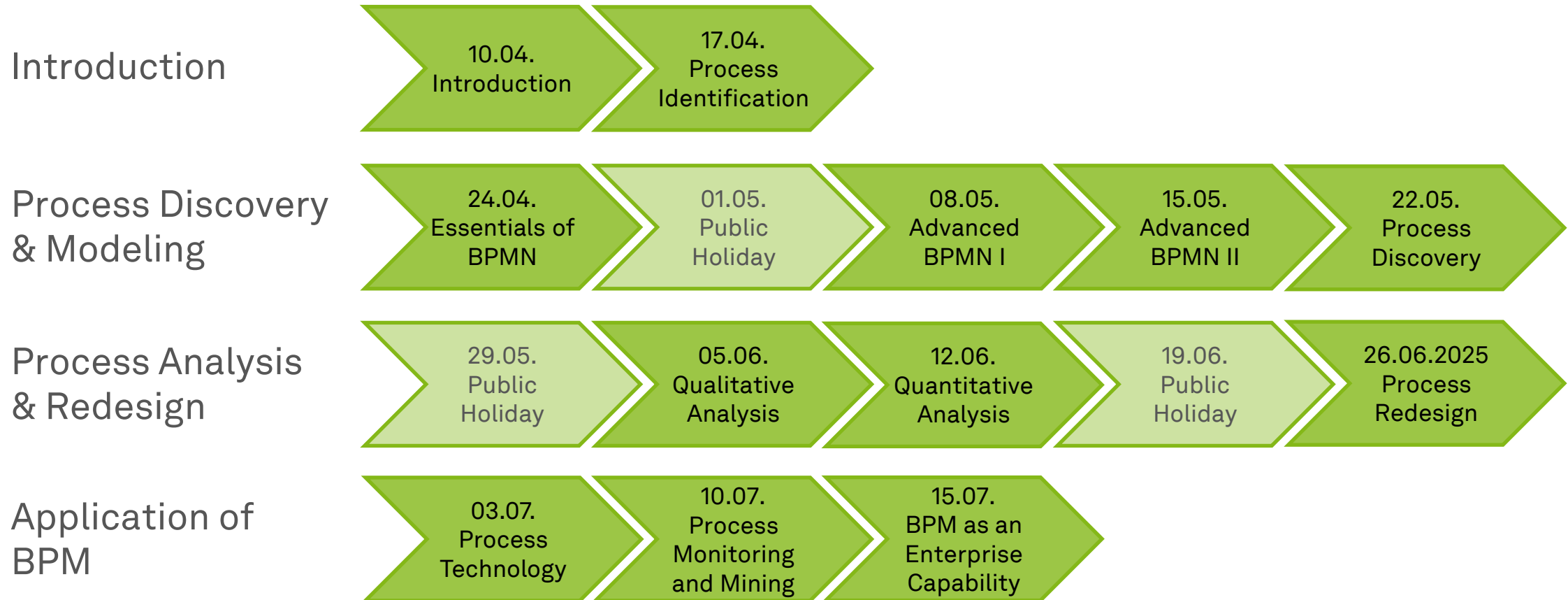
## Questions you should be able to answer after the lecture

- Why do we **need** business processes?
- How do I **structure and model** business processes?
- What techniques can be used to **discover** processes?
- How can I use **qualitative** and **quantitative** process **analysis**?
- How do I **redesign** processes?
- How do I make process models **executable**?
- What does a **BPMS** look like?
- What is **RPA**?
- How can I realize process **monitoring** and **analytics**?
- How can I install BPM as an **enterprise capability**?

# The BPM Lifecycle



# Roadmap





# Anwendungsfach Enterprise Computing (Bachelor)

Pflicht BIS  
(Janiesch)

Pflicht BPM  
(Janiesch)

Pflicht MNP  
(Rehof)

Wahlpflicht EC 8 LP

Wahlpflicht WiWi 15 LP

Fachprojekt EC

Proseminar

Bachelorarbeit

# Unsere Veranstaltungen im neuen Anwendungsfach Enterprise Computing (Bachelor)

Pflicht BIS  
(Janiesch)

Pflicht BPM  
(Janiesch)

Pflicht MNP  
(Rehof)

Wahlpflicht EC 8 LP  
(z.B. 4 ECTS ERCIS Winter School)

Wahlpflicht WiWi 15 LP

Fachprojekt EC –  
Process Automation

Proseminar – EC

Bachelorarbeit

## Mandatory Reading

- Dumas, La Rosa, Mendling & Reijers (2018): Fundamentals of Business Process Management. 2nd Edition. Springer.
  - <https://doi.org/10.1007/978-3-662-56509-4>
- Dumas, La Rosa, Mendling & Reijers (2021): Grundlagen des Geschäftsprozessmanagements. Springer.
  - <https://doi.org/10.1007/978-3-662-58736-2>

## Suggested Further Reading

- **Further Business Process Management Textbooks**

- Becker, Kugeler & Rosemann (2012): Prozessmanagement: Ein Leitfaden zur prozessorientierten Organisationsgestaltung. 7th Edition. Springer. ([LINK](#))
- vom Brocke & Rosemann (2015): Handbook on Business Process Management 1: Introduction, Methods, and Information Systems. 2nd Edition. Springer. ([LINK](#))
- vom Brocke & Rosemann (2015): Handbook on Business Process Management 2: Strategic Alignment, Governance, People and Culture. 2nd Edition. Springer. ([LINK](#))

- **Process Technology**

- Weske (2024): Business Process Management: Concepts, Languages, Architectures. Springer. ([LINK](#))



## Suggested Further Reading

- **Process Mining**

- van der Aalst (2016): Process Mining: Data Science in Action. 2nd Edition. Springer. ([LINK](#))
- van der Aalst & Carmona (2022): Process Mining Handbook. Springer. ([LINK](#))

- **Business Process Management in Practice**

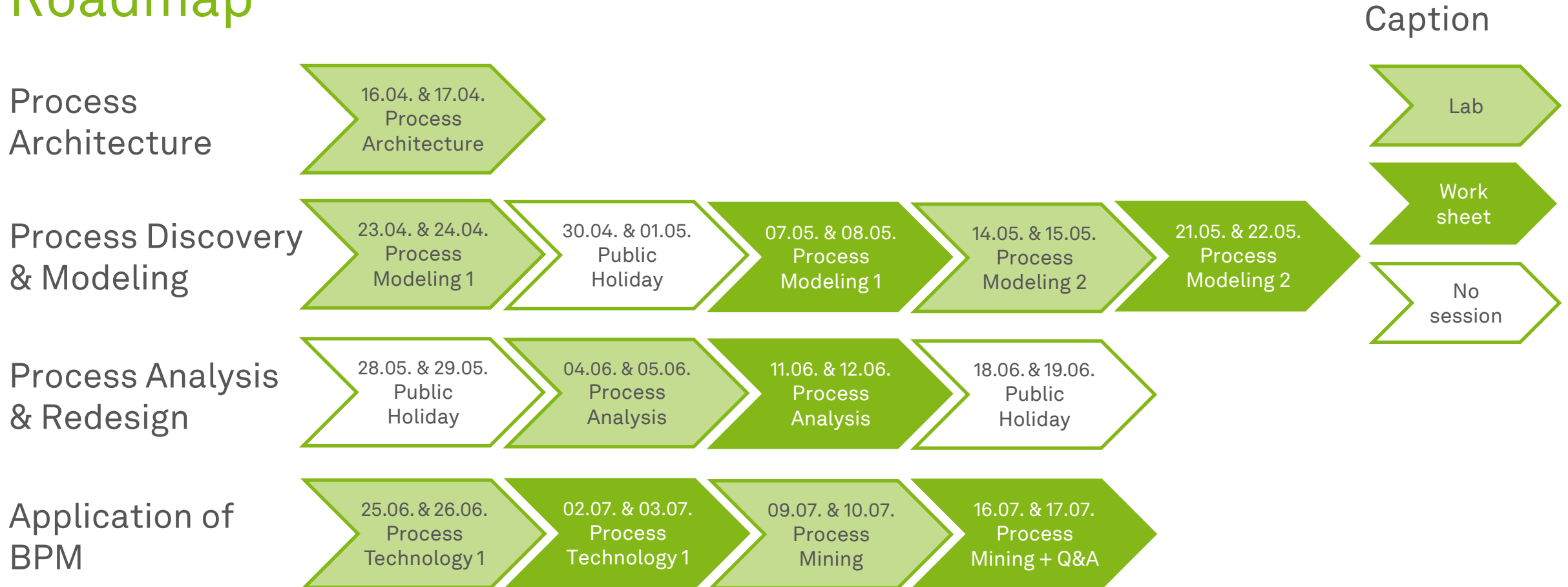
- vom Brocke & Mendling (2018): Business Process Management Cases: Digital Innovation and Business Transformation in Practice. Springer. ([LINK](#))
- vom Brocke, Mendling & Rosemann (2021): Business Process Management Cases Vol. 2: Digital Transformation - Strategy, Processes and Execution. Springer. ([LINK](#))
- vom Brocke, Mendling & Rosemann (2025): Business Process Management Cases Vol. 3: Implementation in Practice. Springer. ([LINK](#))

## Lab Class

- Concurrent to the lecture
  - Tutors **Seyyid A. Ciftci**
  - Lab classes with exercises
- Attend one of two available lab classes
  - Starting from **16.04.2025** and **17.04.2025**
  - Every Wednesday – 14:15 - 15:45h
    - Location: **OH12, 2.063**
  - Every Thursday – 10:15 - 11:45h (if need be)
    - Location: **OH14, 3.04**



# Roadmap



## Assignments

- ...are available at the end of the preceding lab class
- ...must be handed in by **midnight on Tuesday** before the next class
- ...should be submitted in a team of **three**
- ...more next Wednesday and Thursday with Seyyid

- **Example**

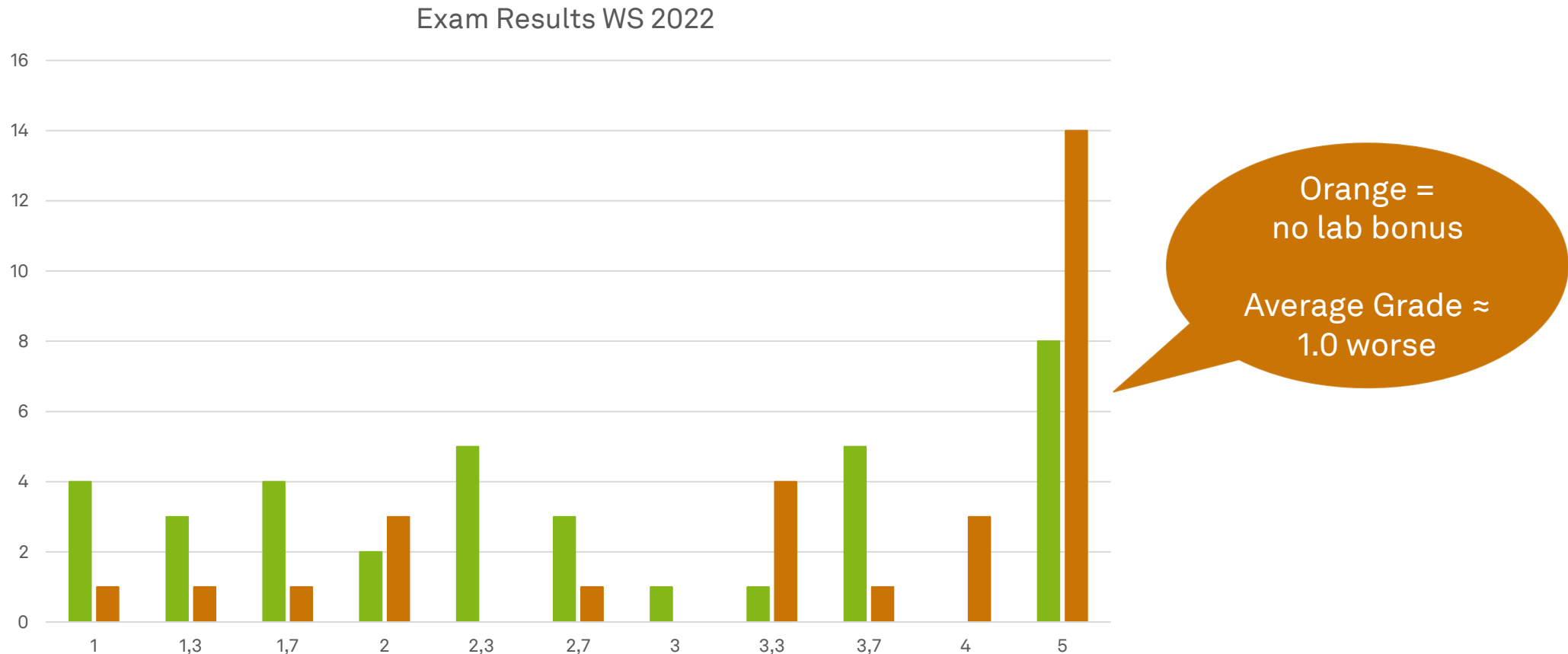
|     |            |                                     |
|-----|------------|-------------------------------------|
| 2   | 23.04.2025 | 2nd Lab Class and 1st Assignment    |
| 3   | 07.05.2025 | 3rd Lab Class and 1st Presentations |
| ... | ...        | ...                                 |

Assignment  
will be online  
after class

Hand in until  
06.05.2025 at  
midnight

Discussion of  
1st assignment

# Reason 1 why you should do your assignments





## Reason 2 why you should do your assignments



You need to **successfully\*** complete  
all assignments to improve your grade of  
the passed exam by one step



# Introduction and Definitions



## Learning Goals of Today

- Get to know the elements and structure of a business process
- Get an initial understanding of business process management and the business process lifecycle
- (Review (in brief) of the history of BPM)

# What is a Process?



Vacation  
Planning



Cleaning plan



Business  
Processes

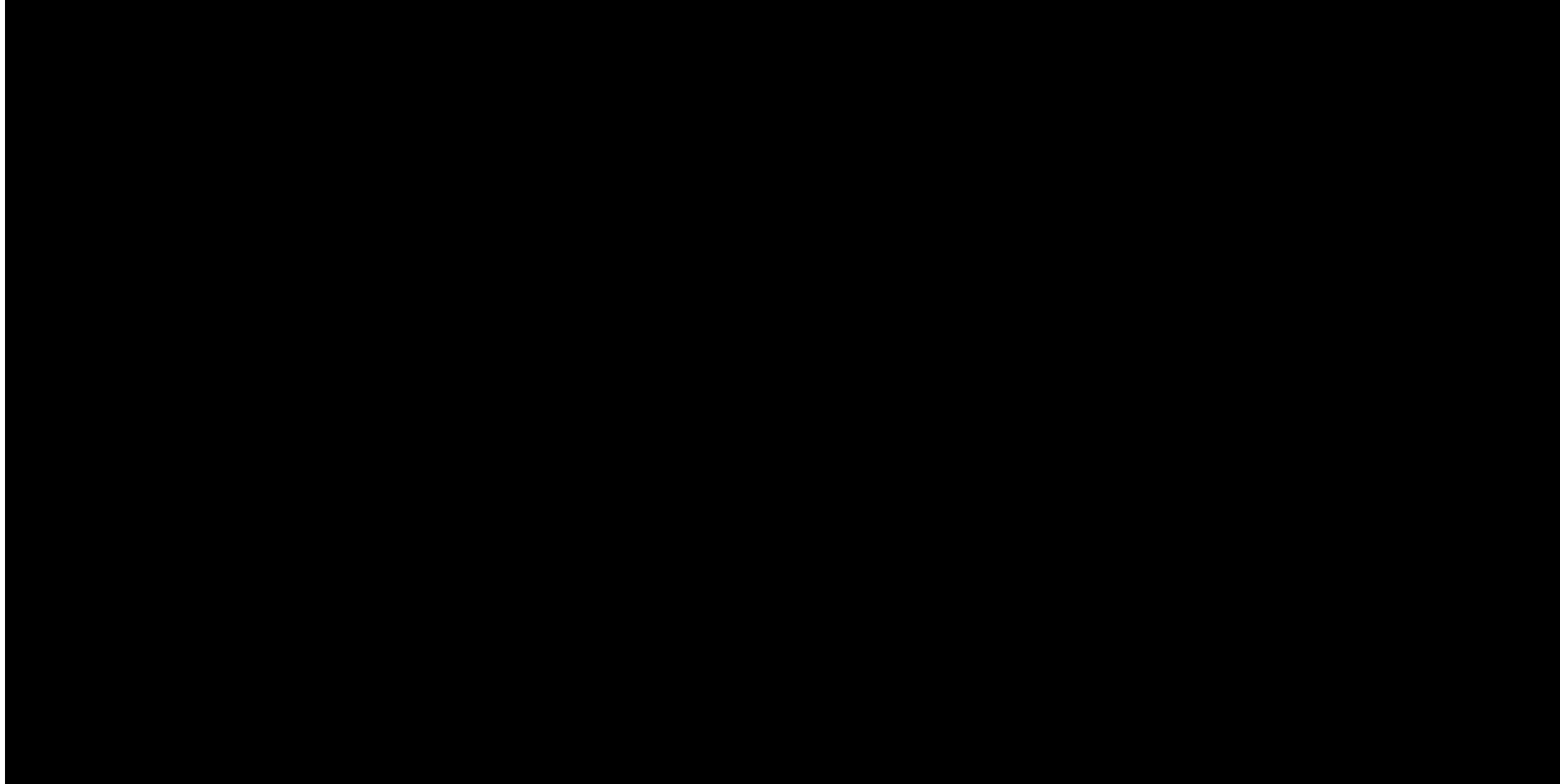


Budget  
Planning



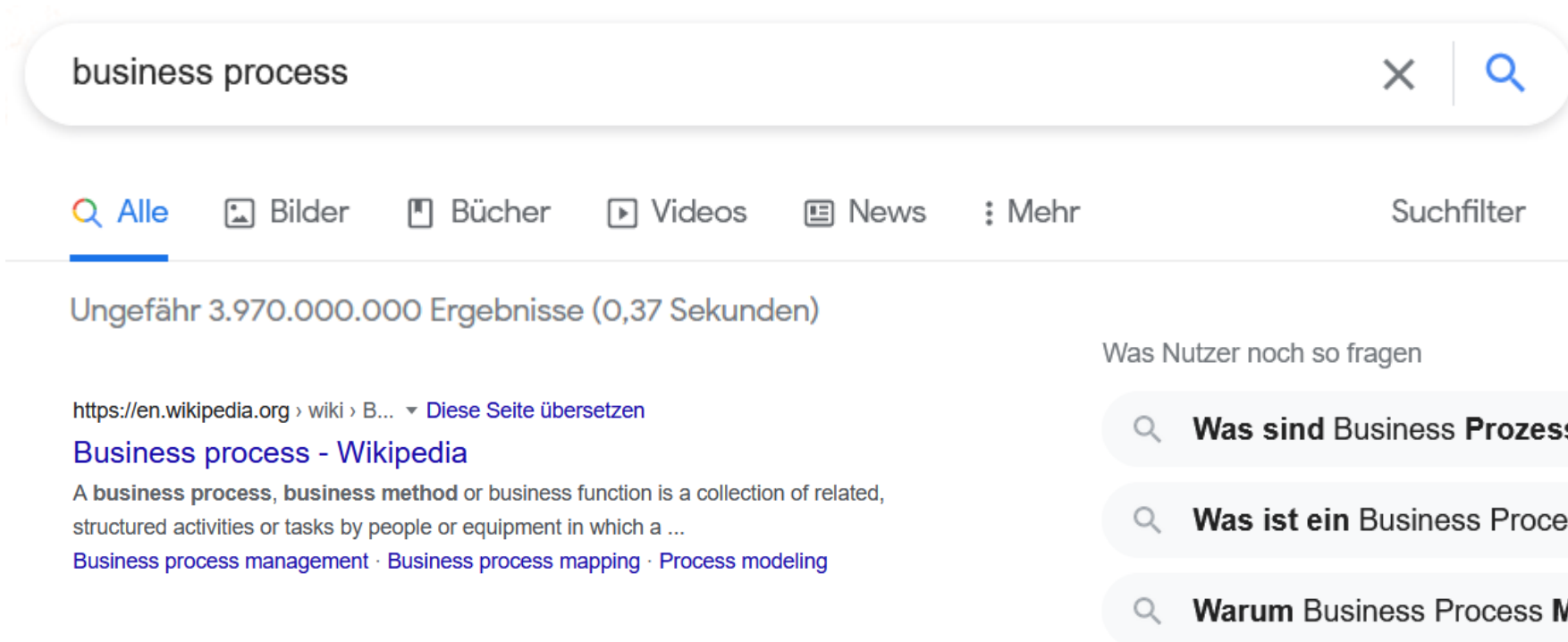
Calendar  
Management

# Business Process Management at Lincoln Trust



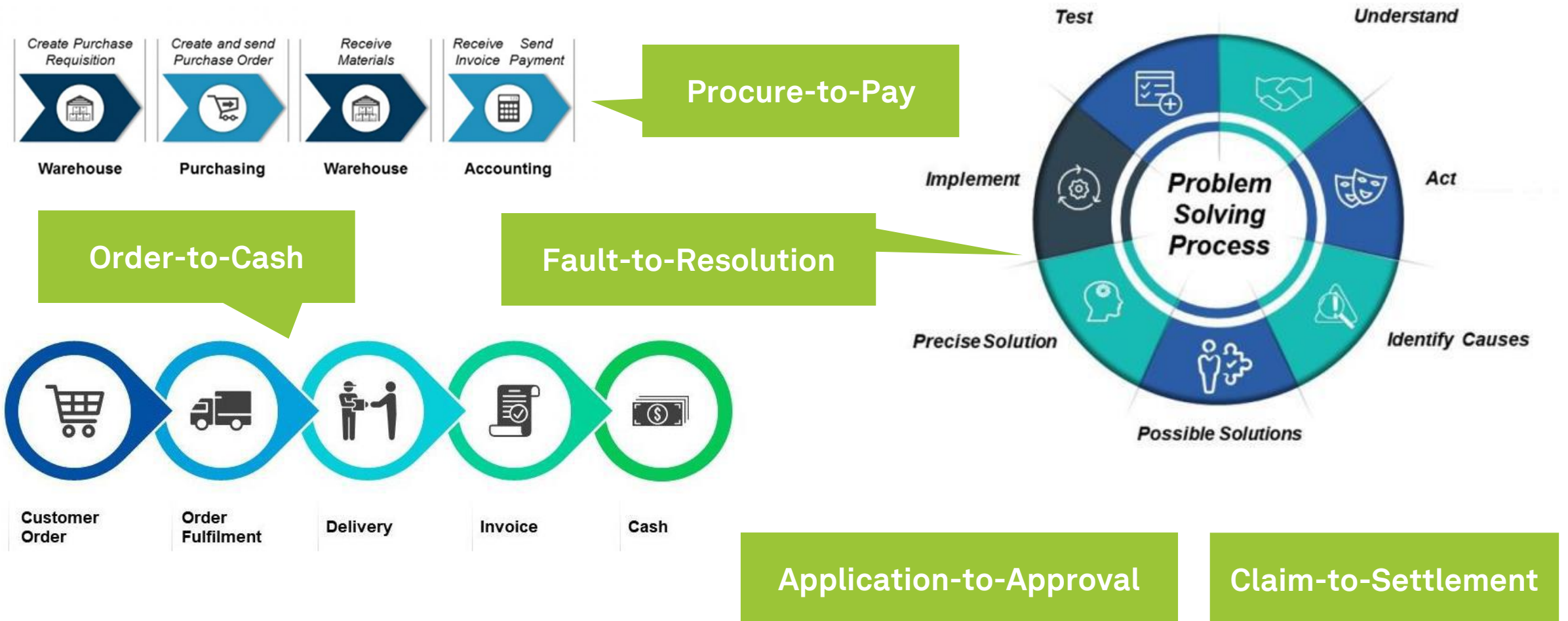
# What is a Business Process?

- Querying Google with “**Business Process**”, the search engine delivers about **4 billion results**





# Examples for Business Processes



# Elements and Structure of a Process

- Business processes are defined by a set of **"ingredients"**
  - **Actors** (internal, external // humans, organizations, software systems)
  - **Tasks and Activities** (single units of work // more coarse-grained units of work)
  - **Events** (things that happen instantaneously without resources)
  - **Decision Points** (decision affecting the way the process is executed)
  - **Physical Objects** (equipment, materials, products, paper documents)
  - **Informational Objects** (electronic documents, electronic records)
- Business Process have an **outcome** for the consumer that can be
  - **Positive** or
  - **Negative**

# Elements and Structure of a Process

- **Actors**

- Entities that are performing process activities and generate events
- Can be **humans**, **organizations**, or **software systems**,
  - For example, “*Airport Security Officer*”, “*Financial Officer*”, etc. (roles)
  - “ERP”, “CRM”, etc. (software systems)



# Elements and Structure of a Process

- **Activities**

- Active elements, for example “Check Luggage”
  - **Time-consuming**
  - **Resource-demanding**
  - **State-changing**



# Elements and Structure of a Process

- **Events**

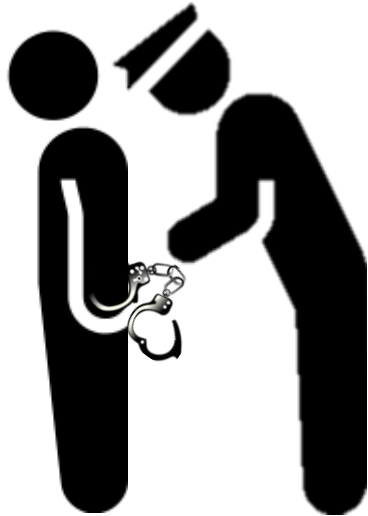
- Passive elements, for example “Sharp Object Found”
- Stands for conditions or circumstances
- Atomic and instantaneous



# Elements and Structure of a Process

- **Decision Points**

- Affect future process execution
- For example, decide “whether the offense was punishable”



- **Objects**

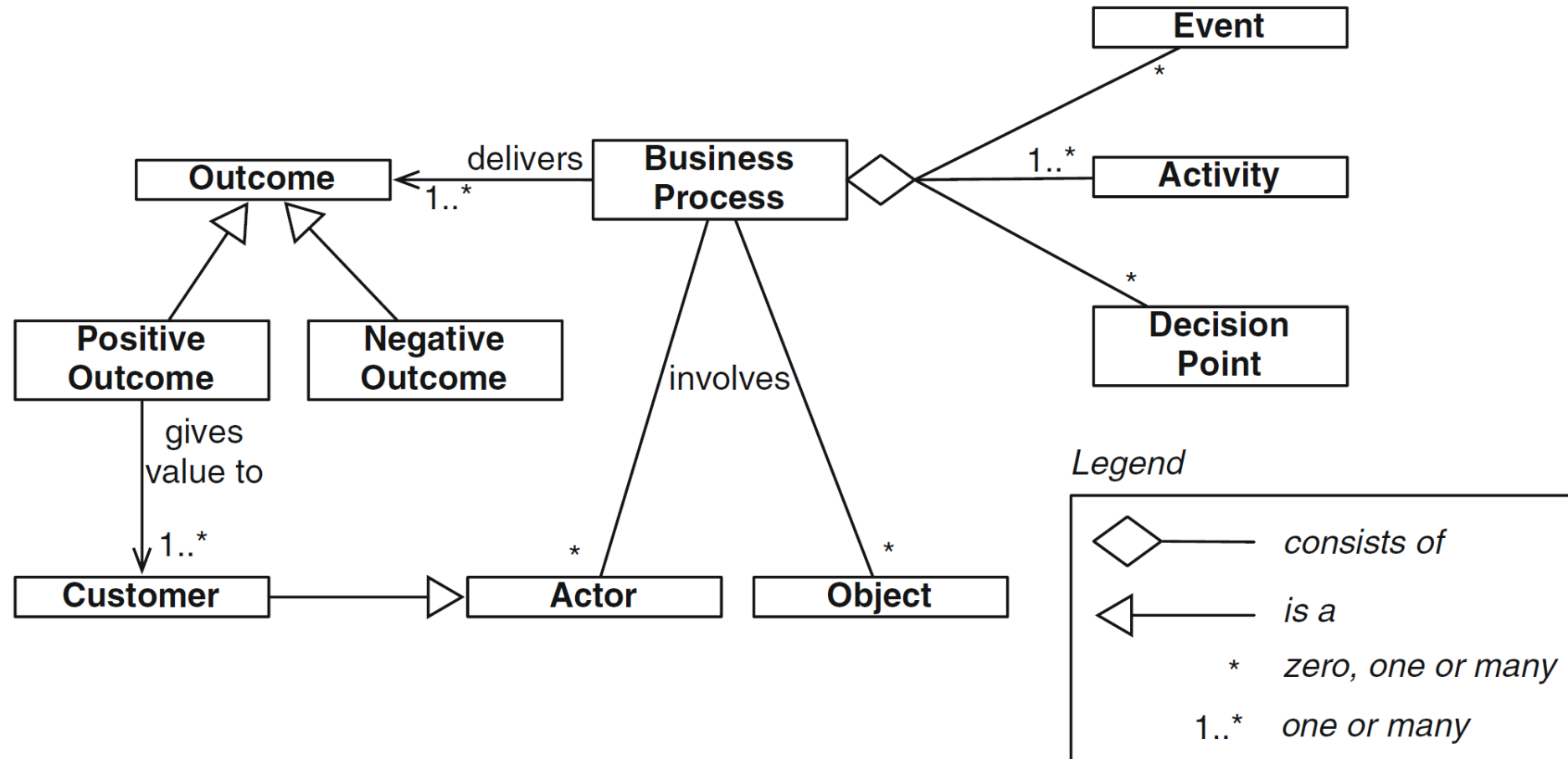
- For example, warrant, sales order, digital object, etc.
- Can undergo **state changes**
- Contain physical or electronic information



Object



# Elements and Structure of a Process Diagram



# Definition of a Business Process

- A business process is
  - a **collection** of inter-related events, activities, and decision points
  - that involve a number of **actors and objects**
  - which collectively **lead to an outcome that is of value** to at least one customer
- The activities of a business process **jointly** realize a business goal



So then, what is the M  
in **BPM**?

# An Abstract Definition of Business Process Management

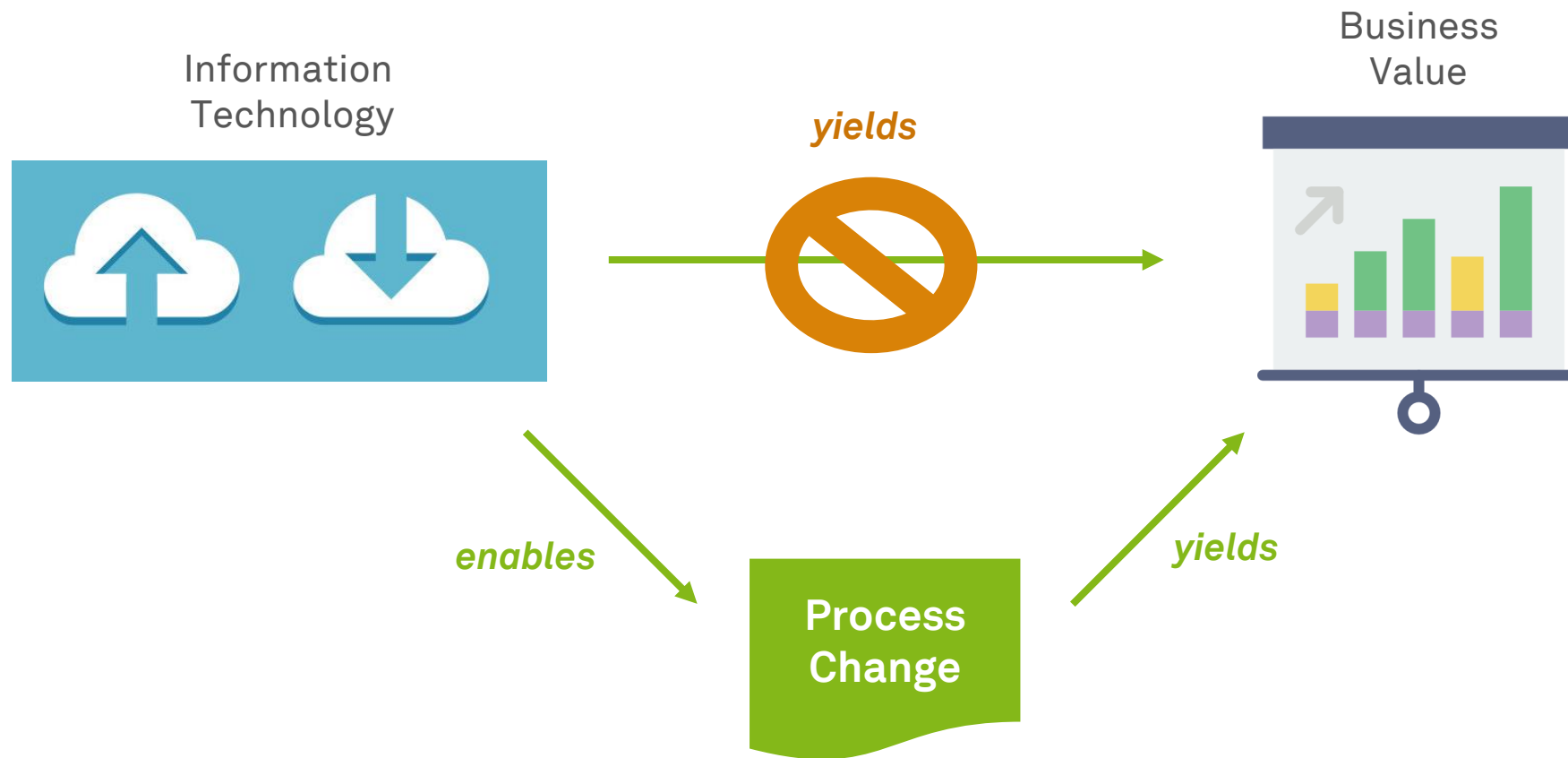
- Business Process Management (BPM) is the **art and science** of overseeing how work is performed in an organization
  - to **ensure consistent outcomes** and
  - to **take advantage of improvement\* opportunities**
- \* improvement: one-off or continuous
  - reduce costs
  - reduce execution times
  - reducing error rates
  - **gain competitive advantage through innovation**



# Definition of Business Process Management

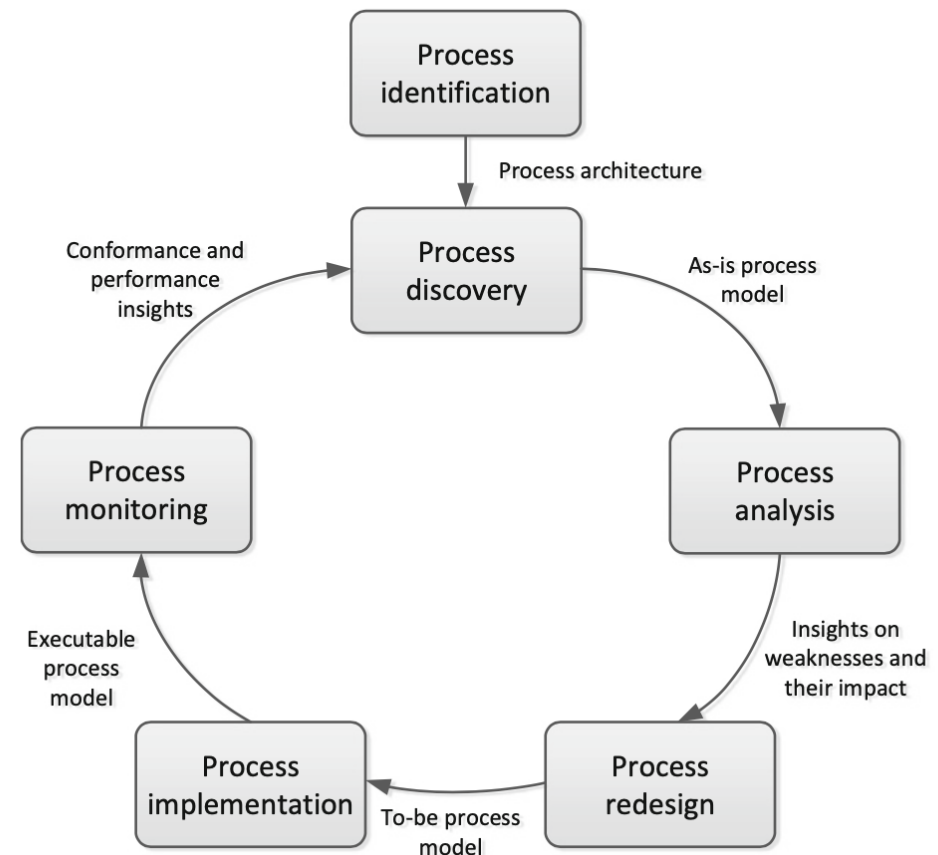
- Business Process Management (BPM) is
  - a **body of** methods, techniques, and tools
  - to **identify, discover, analyze, redesign, execute**, and **monitor** business processes
  - in order to **optimize** their performance.
- BPM **is...**
  - enabling technical and organizational improvements of processes
  - providing powerful methods for documenting processes
  - providing frameworks for introducing new processes
- BPM **is not ...**
  - the automation of manual tasks
  - re-engineering the enterprise
  - change management

# Why Do We Need BPM?



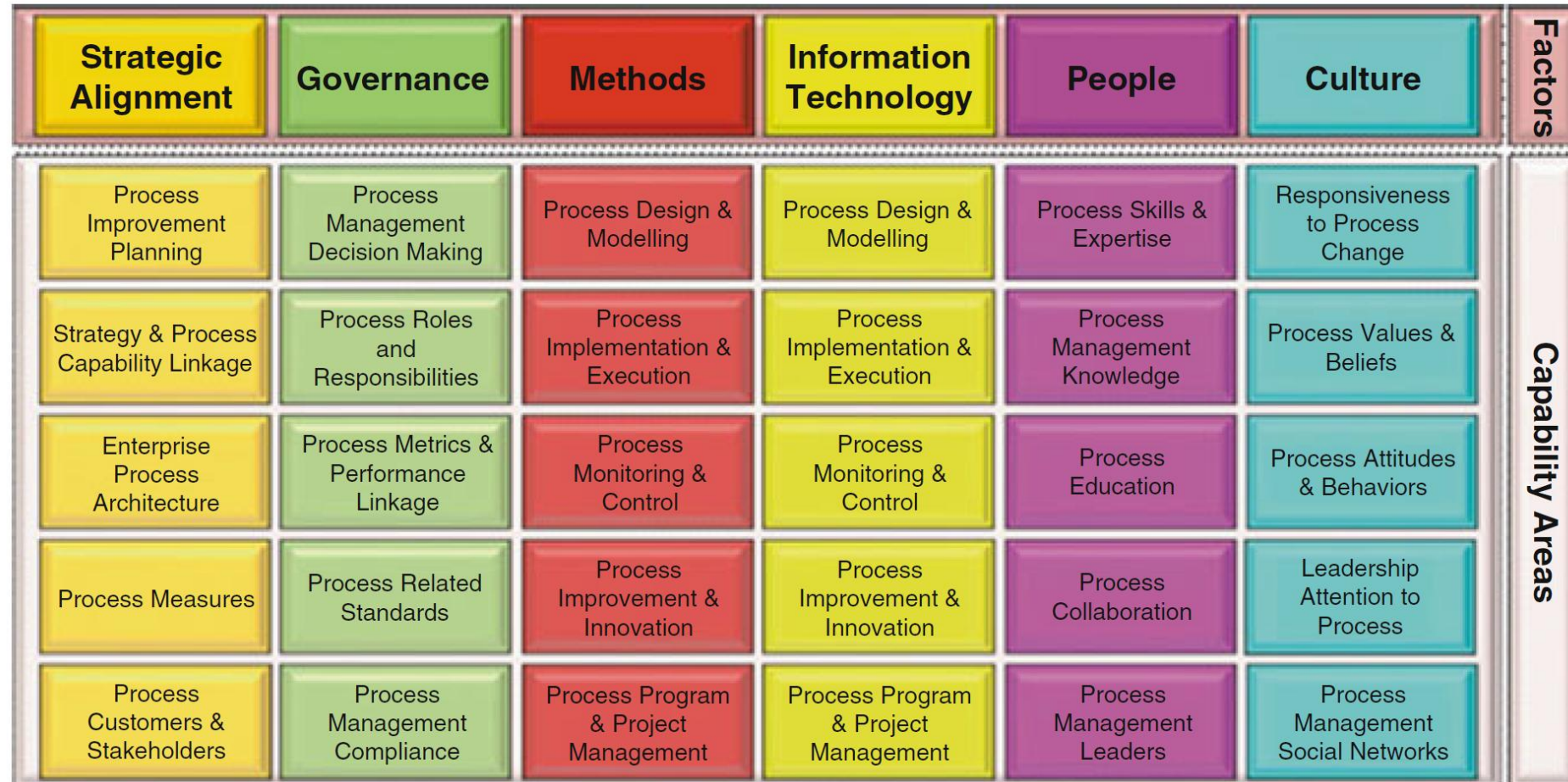
# BPM Lifecycle

- Consists of **phases** related to each other
- Organized in a **cyclical structure**, showing their logical dependencies
- **No strict temporal ordering**, some phases might occur during each of the phases





# BPM Six Core Elements



Rosemann & vom Brocke (2015)

# A Brief History of Business Process Management



# From Function Orientation to Process Orientation

## Function Orientation

- Organizations seek **local** optimization and perfection of functional areas
- The aim is **to increase quality**
- Technological and organizational developments
  - Outsourcing (e.g., call centers)
  - Integrated standard software (ERP)

## Process Orientation

- **Focus on processes** within the organization – not on functional areas or local optimization!
- Assume that running a business is a **continuous process**
- It makes sense to structure an organization along its **technical processes**

# From Function Orientation to Process Orientation

## Function Orientation

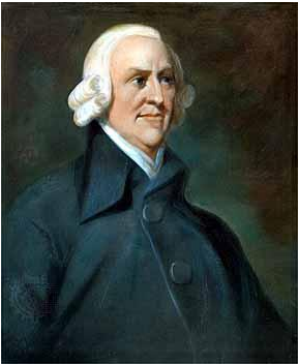
- Organizations seek **local** optimization of functional areas
- The focus is on **Problems**
  - Autonomous functional areas
  - Cost increase through internal arrangements and coordination
- Technical developments
  - Outsourcing (e.g., call centers)
  - Integrated standard software (ERP)

## Process Orientation

- **Focus on processes** within the organization. Functional areas are **Solution**
  - Organizations need to move from functional orientation to **process orientation**
- Assessment of a process is a **co**
- It makes sense to structure an organization along its **technical processes**

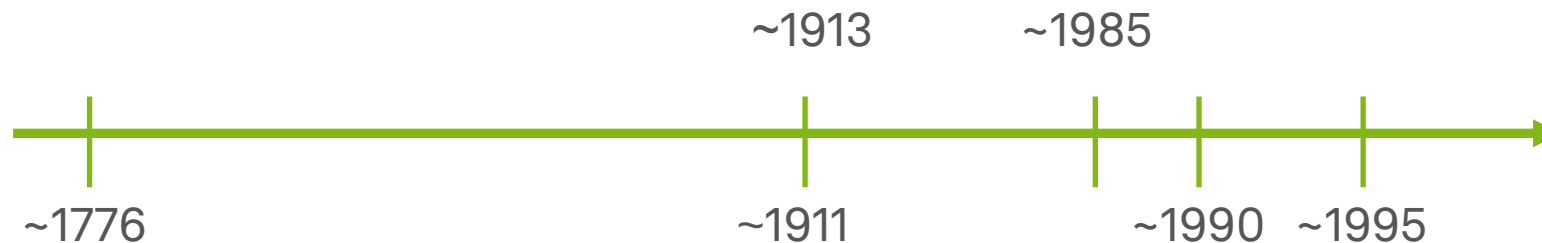
# History of BPM

## Management Concepts (1776)



**Adam Smith**  
1723 - 1790

- Book “**An Inquiry into the Nature and Causes of the Wealth of Nations**”
- **Specialization** of labor
- **Organized** around tasks
- **Observation** that specialized workers could produce far more output than the same number of generalists
- Gives example of pin maker

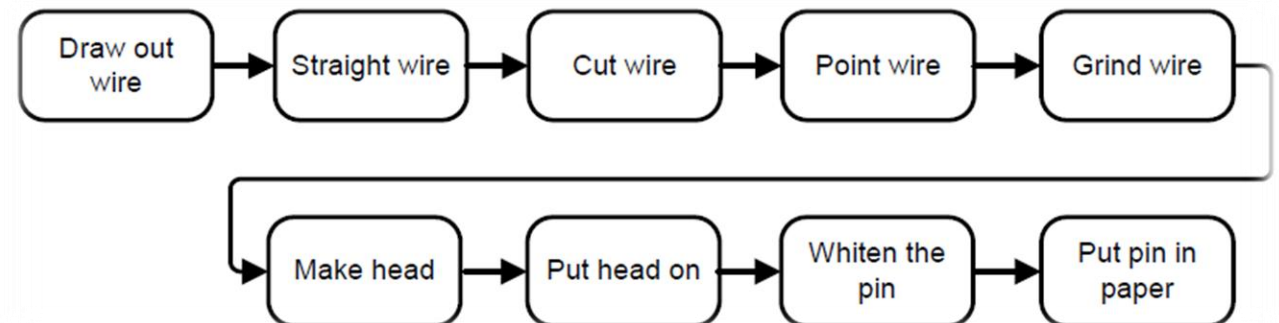
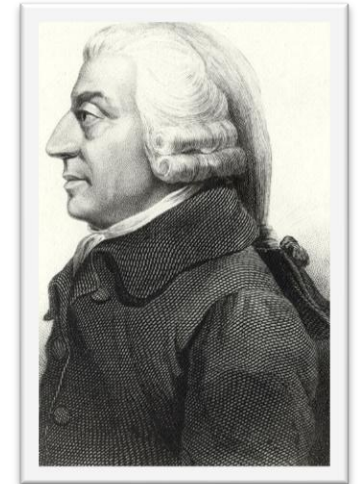


Dumas et al. (2018), Rosemann (2002)



# Adam Smith: Processes and Specialization of Labor

- “To take an example, the trade of a pin-maker: But in the way in which this business is now carried on, it is divided into a number of branches:
- One man draws out the wire; another straightens it;
- a third cuts it; a fourth points it; a fifth grinds it at the top for receiving the head;
- to make the head requires three operations; to put it on is a peculiar business; to whiten the pins is another; to put them into the paper;
- and the important business of making a pin is, in this manner, divided into about eighteen distinct operations.”

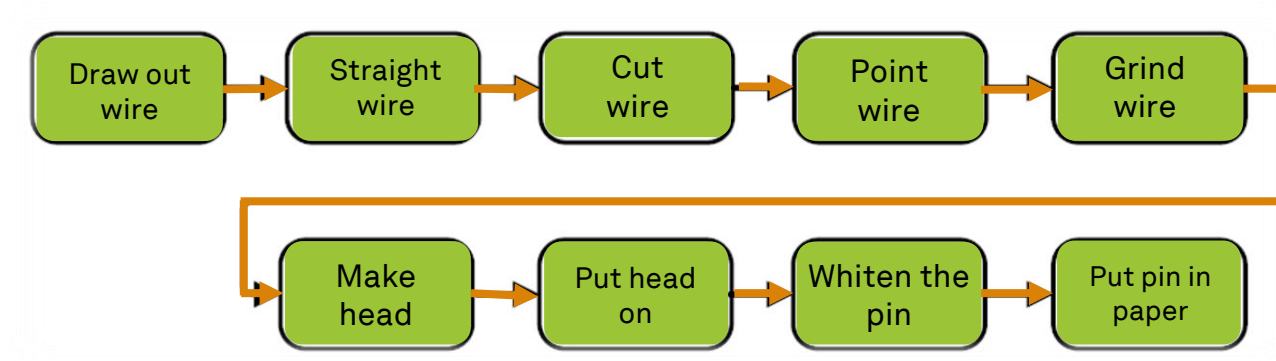


Dumas et al. (2018), Hansen et al. (2019)

# Task Efficiency and Coordination Efficiency

## Task Efficiency

## Coordination Efficiency





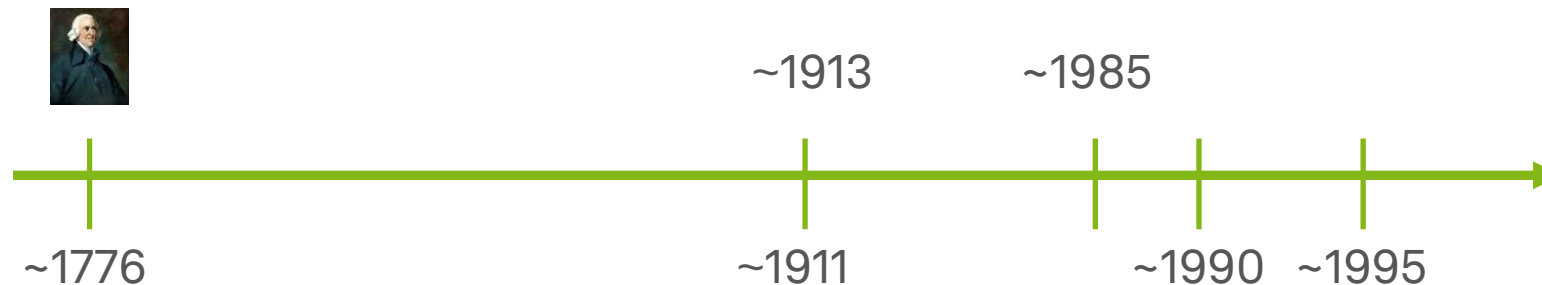
# History of BPM

## Management Concepts (1911)



**Frederick Taylor**  
1856 - 1915

- **Scientific Management** (Taylorism)
  - Analyzes and synthesizes workflows to improve economic efficiency
  - Introduction of managers
  - **Assumption** that behavior at work can be engineered and designed according to principles of rationality and efficiency



Dumas et al. (2018), Rosemann (2002)

# History of BPM

## Management Concepts (1913)



Henry Ford  
1863 - 1947

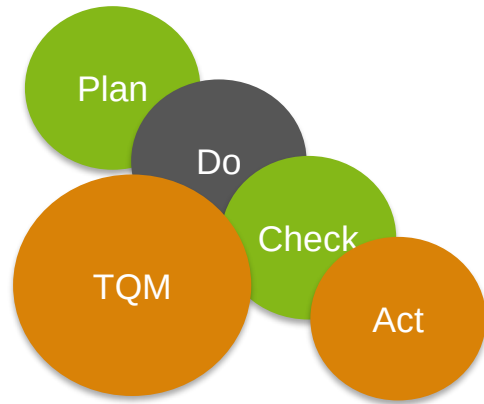
- Introduction of a moving **assembly line**
- Bring the work to the worker – not vice versa
- **Alfred Sloan**: Adapted the idea of Smith to management and created smaller divisions aiming at a division of professional labor



Dumas et al. (2018), Rosemann (2002)

# History of BPM

## Management Concepts (1985)



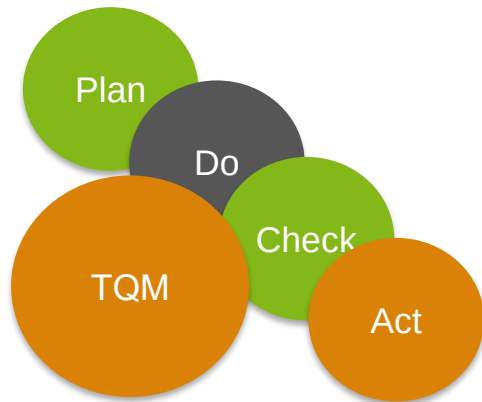
Total Quality  
Management  
~1985

- There are many different origins of quality management
  - **Frederick Taylor** laid a foundation for quality management by empowering the efficiency movement
  - **Walter Shewhart** created a method for quality control for production (1924 firstly proposed)
  - **William Deming** brought Shewhart ideas to Japan in the 1950s



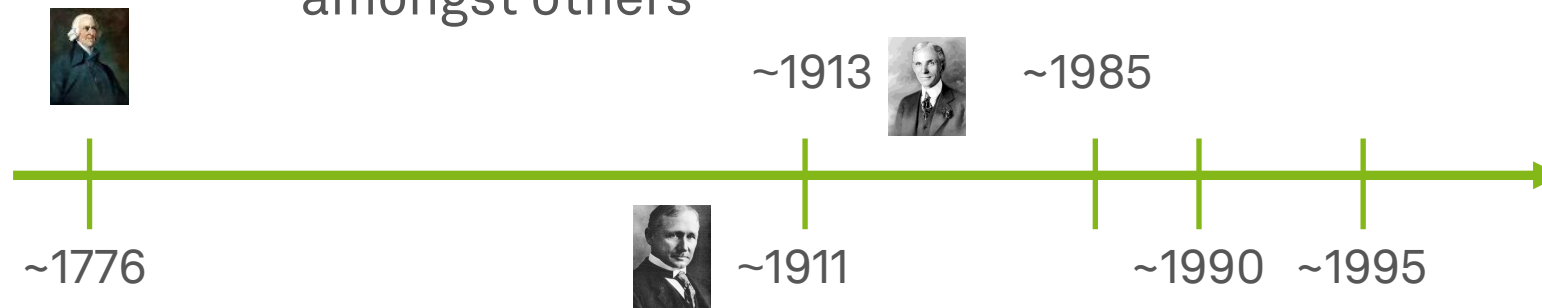
# History of BPM

## Total Quality Management



Total Quality  
Management  
~1985

- **Organization-wide** efforts to continuously improve quality of products and services
- Necessity of **ongoing** improvement efforts
- There is **no single** popular and widely agreed-upon **approach**
  - Effort heavily rely on tools and **techniques** of quality control
  - Notable **approaches** include **lean management** and **six sigma** amongst others



# History of BPM

## Management Concepts (1990)

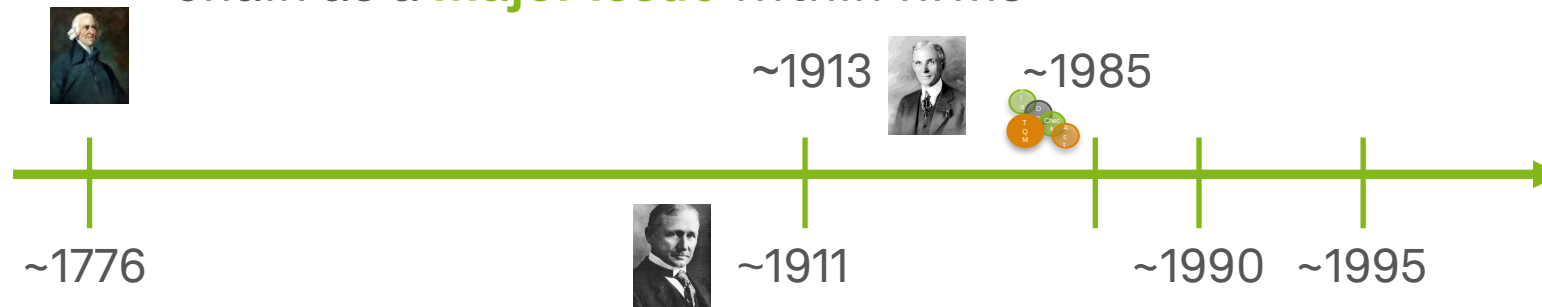


Business Process  
Reengineering  
~1990

*“Sound strategy  
starts with having  
the right goal.”*

Michael Porter

- The basic idea of process orientation goes back to **Porter’s value chain**
  - Added value is created within along the **value chain** of a business (i.e., cross-functional)
  - Decomposing the value chain exposures **transactions**, which can be assembled to **processes**
  - Porter further stresses the concept of **interoperability** across the value chain as a **major issue** within firms



Porter (2000), Hammer & Champy (1993)

# History of BPM

## Management Concepts (1990)



Business Process  
Reengineering  
~1990

- Davenport and Short, as well as Hammer and Champy and others, brought **reengineering** of processes into play
- In 1990, they popularized the use of Process Reengineering as **Business Process Reengineering** or Business Process Redesign (BPR)

*“Reengineering means to **disregard all the assumptions and traditions** of the way business has always been done, and instead develop a **new, process-centered business organization** that achieves a **quantum leap forward** in performance.”*

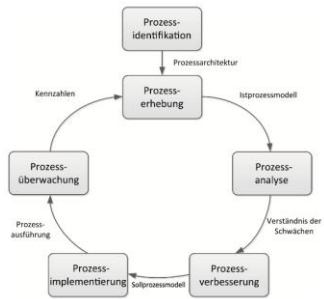
Michael Hammer



Hammer & Champy (1993)

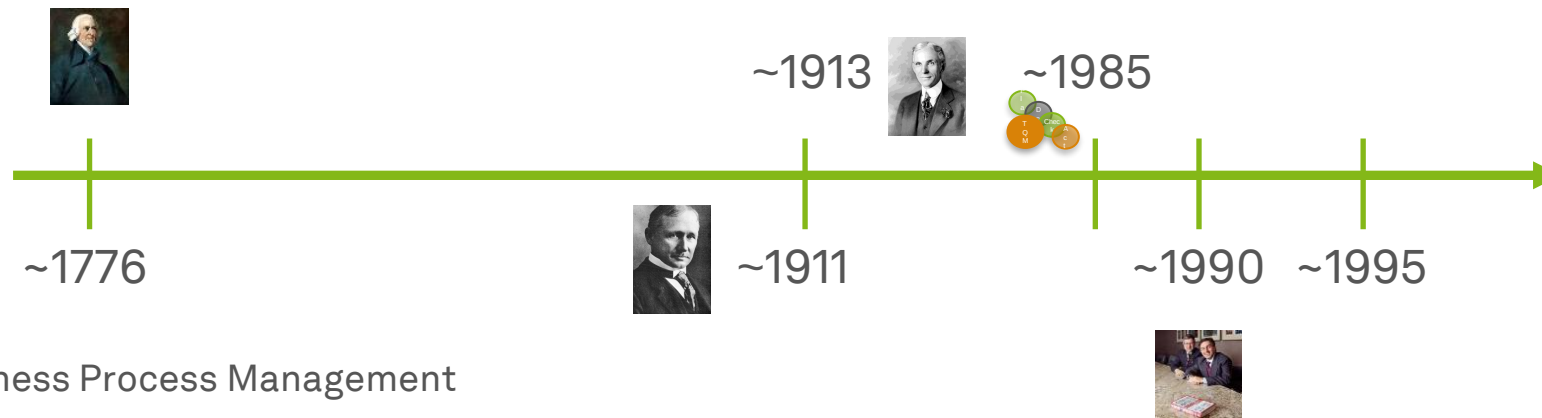
# History of BPM

## Management Concepts (1995+)



**Business Process Management**  
~1995

- The emergence of **Business Process Management** was caused due to **dissatisfaction and risks** with BPR
  1. Many organizations **misused** the concept of **BPR** to any possible project that did not contain processes at its core
  2. Redesign of processes has often been progressed **too radical**
  3. There was **no sufficient (IT) support** on BPR projects

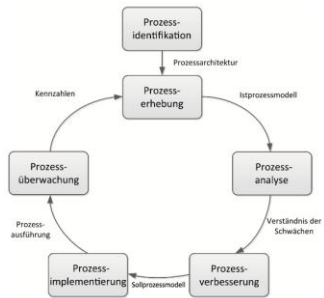




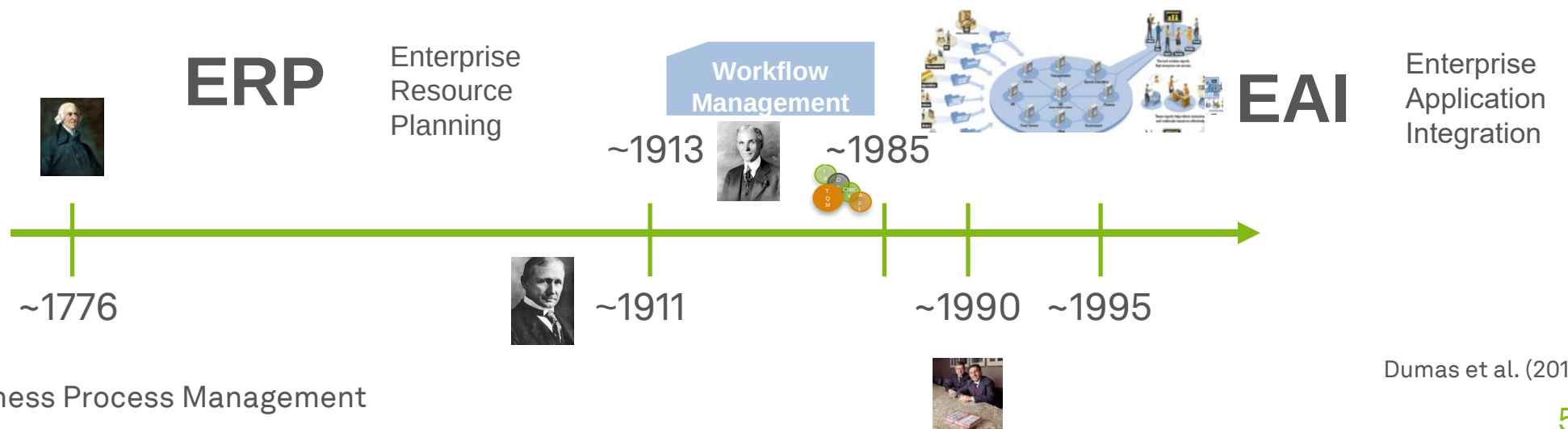
# History of BPM

## Management Concepts (1995+)

- The rise of **process orientation**
  - **Studies** show that process-orientated organizations did better than non-process-oriented organizations
  - Different types of **IT systems emerged** made it easier to implement changes to business processes



**Business Process Management**  
~1995



## Learning Goals of Today

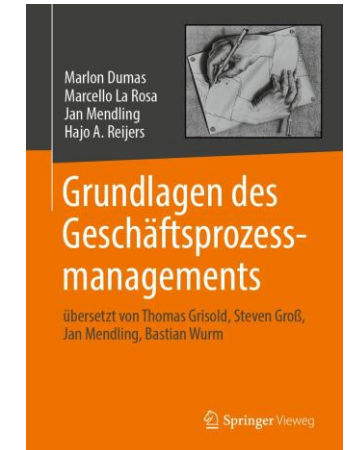
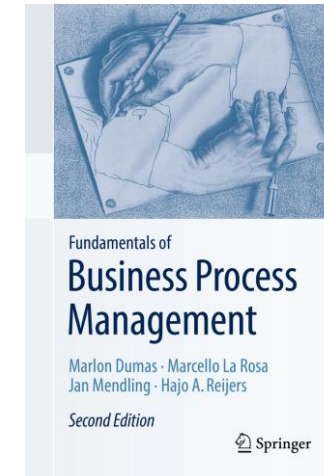
- Get to know the elements and structure of a business process
- Get an initial understanding of business process management and the business process lifecycle
- (Review (in brief) of the history of BPM)

## Recap Questions

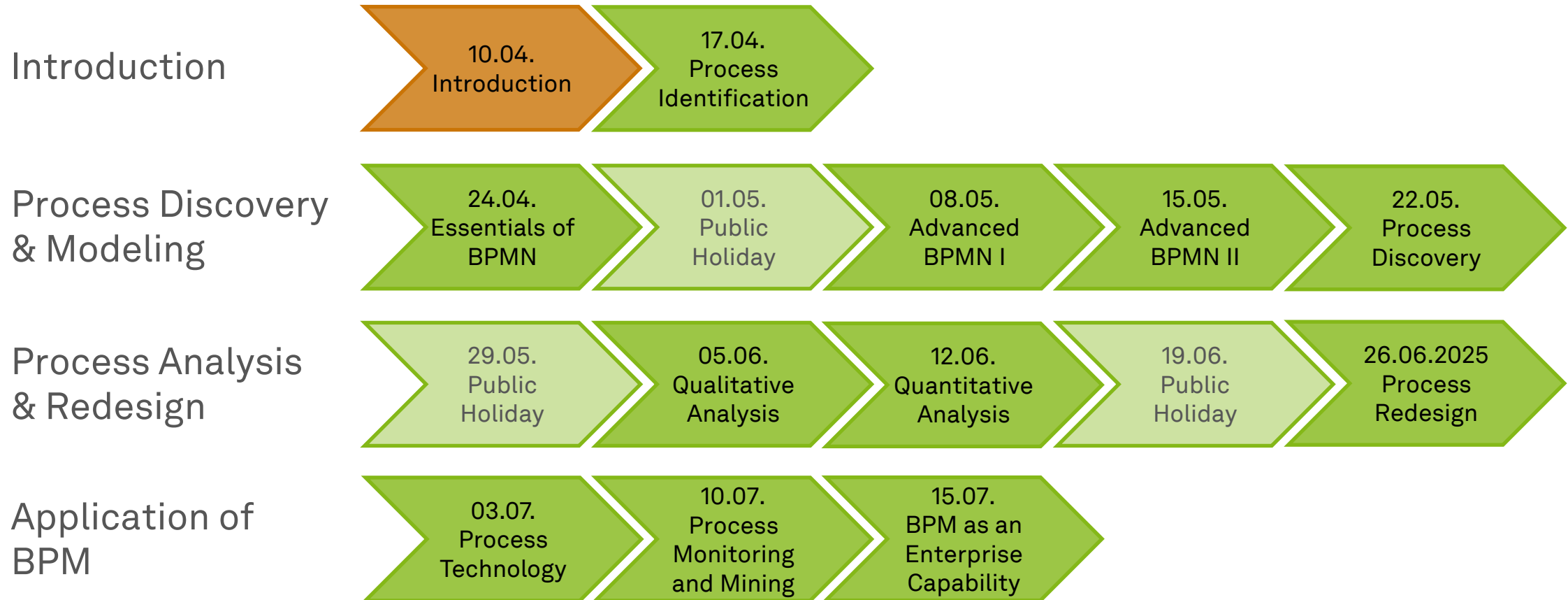
- What is a business process and how is BPM defined?
- Why do we need BPM to yield business value from information technology?
- (What are the differences between function orientation and process orientation?)
- (What are the major steps in the evolution of the specialization of labor to BPM as we know it today?)

## Literature

- **Section 1:** Introduction to Business Process Management



# Roadmap



# Any Questions about BPM?







[ec.cs.tu-dortmund.de](https://ec.cs.tu-dortmund.de)

